

CONFIDENTIAL — INTERNAL DOCUMENT

# TORC

Roadside Assistance Platform

Complete Features & Capabilities Document

**3**

Applications

**75+**

Features

**23**

Database Tables

**7**

Email Templates

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# 1. Platform Overview

Architecture, technology stack, and application summary

3

Applications

79

Total Pages

48

DB Migrations

2

Native Platforms

## Technology Stack

| Layer              | Technology                                                        |
|--------------------|-------------------------------------------------------------------|
| Frontend           | React 18 + TypeScript, Tailwind CSS, Framer Motion, Vite          |
| Native Wrapper     | Capacitor (iOS + Android) — wraps React SPA as native app         |
| Backend / DB       | Supabase (PostgreSQL 15, Auth, Storage, Edge Functions, Realtime) |
| Payments           | Stripe (PaymentIntents, PaymentMethods, Customer objects)         |
| Email              | Resend API via Supabase Edge Function                             |
| Maps               | Google Maps JavaScript API (Places, Directions, Geocoding)        |
| Real-time          | Supabase Broadcast Channels, Postgres Changes, Presence           |
| Push Notifications | Native APNS (iOS) / FCM (Android) via Capacitor                   |
| Chat Encryption    | AES-256-GCM with PBKDF2-derived keys                              |

## Three Applications

**Customer App (iOS + Android)**

Request roadside assistance, track providers in real-time, pay securely, rate service. 25 core pages covering the full service lifecycle from request to completion.

Capacitor

Stripe Payments

Live GPS Tracking

**Admin Dashboard (Web)**

Full platform management — users, providers, jobs, payments, payouts, support tickets, analytics, settings, team, and audit trail. 24 pages with real-time monitoring.

Admin Only

Analytics

Live Dispatch

**Provider App (iOS + Android)**

Accept job requests, navigate to customers, complete services, earn money, manage documents and payouts. 30 pages covering onboarding through active job execution.

Capacitor

Job Dispatch

Earnings Dashboard

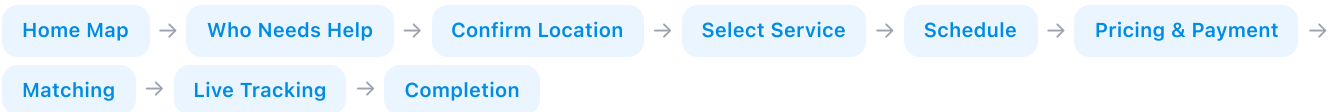
## 2. Customer App Features

25 pages — Full service request, tracking, payment, and account management

### Authentication & Onboarding (6 screens)

| Screen          | Features                                                                       |
|-----------------|--------------------------------------------------------------------------------|
| Splash          | Auto-redirects based on auth state; crash recovery checks for in-progress jobs |
| Login           | Email/password auth; password visibility toggle; forgot password link          |
| Signup          | Name, email, phone, password; terms acceptance; welcome email sent on creation |
| Verify Email    | OTP/link verification; resend option; pending email display                    |
| Forgot Password | Email-based password reset request                                             |
| Reset Password  | New password entry with confirmation and validation                            |

### Service Request Flow



| Step              | Key Features                                                                                                                                                                                 |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Home Map          | Interactive Google Map; current location; quick stats (saves, rating, member since); active job banner for crash recovery; real-time job polling every 8s; auto-cleanup of stale jobs (>12h) |
| Who Needs Help    | Select "Myself" or "Someone Else"; add name + phone for third party                                                                                                                          |
| Confirm Location  | Draggable map marker; Google Autocomplete address search; hazard location toggle (adds fee); current GPS auto-fill                                                                           |
| Service Selection | Grid of available services; search/filter; icon, name, description, base price; loads from Supabase services table                                                                           |
| Schedule Service  | Date/time picker for future bookings; scheduling fee applied; toggle immediate vs. scheduled                                                                                                 |
| Pricing & Payment | Full cost breakdown (base + service fee + hazard fee + scheduling fee + tax); Stripe card integration; add/select saved cards; set default payment method                                    |
| Matching          | Tiered provider dispatch (5mi → 15mi → 30mi); Supabase broadcast to eligible providers; ETA calculation; cancel with reason; provider acceptance detection                                   |

|                           |                                                                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Live Tracking</b>      | Real-time GPS map; provider route rendering via Google Directions; status progression (enroute → arrived → working → complete); encrypted chat; phone calling; share tracking link; 911 emergency button; cancel support |
| <b>Service Completion</b> | 1-5 star rating; optional text feedback; tip selection (\$0-\$20); before/after photo capture; cost summary; email receipt sent                                                                                          |

## Activity, Messages & Explore

| Feature                | Details                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Activity</b>        | Dual tabs: Upcoming / Past; infinite scroll pagination; status badges; click to track or view detail                                                |
| <b>Service History</b> | Filters: All, Completed, Cancelled; expandable detail rows; CSV receipt download                                                                    |
| <b>Messages</b>        | All active job conversations; search by provider name; unread badge; encrypted chat modal; privacy name formatting                                  |
| <b>Explore</b>         | Google Places API — nearby auto shops, gas stations, car washes; map + list views; distance filter; shop detail modal with hours, phone, directions |
| <b>Notifications</b>   | Filter: All / Unread; color-coded by type; mark read; action URL deep links                                                                         |

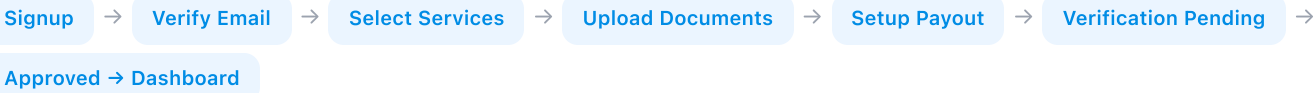
## Profile & Account Management

| Page                    | Capabilities                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------------|
| <b>Profile Hub</b>      | Avatar, name, stats; menu navigation to all settings; dark/light theme toggle; logout            |
| <b>Personal Info</b>    | Edit first name, last name, phone; auto-save to Supabase                                         |
| <b>Vehicles</b>         | Add/edit/delete vehicles; year, make, model, plate, VIN, color; set default vehicle              |
| <b>Payment Methods</b>  | View saved cards (brand, last 4, expiry); add new via Stripe; set default; delete                |
| <b>Account Security</b> | Change password; account deletion with confirmation                                              |
| <b>Help Center</b>      | 5 categories with expandable FAQ articles; support ticket creation with priority; ticket replies |
| <b>Reporting</b>        | Job counts, spending analytics, date range filters; CSV export of all job data                   |

### 3. Provider App Features

30 pages — Onboarding, job execution, earnings, documents, and profile management

#### Onboarding Flow



| Step                 | Details                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Provider Signup      | Name, email, phone, password; auto-creates profiles + provider_profiles rows; welcome email sent                                                                                                                |
| Service Selection    | Multi-select: Towing, Jump Start, Lockout, Fuel Delivery, Tire Change, Winch, Minor Repair, Diagnostic, EV Charge, Motorcycle, Consultation, Emergency Help                                                     |
| Document Upload      | Driver's License, Insurance, Vehicle Registration, Towing Credentials; expiry date picker; status tracking (pending/approved/rejected); rejection reason display; "Documents Under Review" email sent on submit |
| Payout Setup         | Bank Account, PayPal, or Venmo; multiple methods with default selection; encrypted storage                                                                                                                      |
| Verification Pending | Polls every 8s for approval; real-time listener; auto-navigates to dashboard on approval; shows rejection reasons                                                                                               |

#### Dashboard & Job Flow

| Feature        | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Home Dashboard | Online/offline toggle; today's earnings; completed jobs count; provider rating; expired document warnings; incoming job alerts with 30s countdown + audio/haptic                                                                                                                                                                                                                                                                                                                                                                 |
| Job Request    | Full-screen with service type, pickup address, customer name, distance, pricing breakdown; 30s countdown timer with color urgency; accept/decline actions                                                                                                                                                                                                                                                                                                                                                                        |
| Active Job     | <b>Status flow:</b> En Route → Arrived → Working → Photos → Complete<br><b>Map:</b> Live GPS of both parties; Google Directions route rendering<br><b>Communication:</b> Encrypted chat with image upload; WebRTC audio calling; unread message badge<br><b>Controls:</b> Navigate (turn-by-turn); I've Arrived (auto-triggers at 0.5mi); Start Service; Complete Job<br><b>Photos:</b> At least 1 completion photo required via native camera<br><b>Cancel:</b> With reason; atomic RPC; customer notified via broadcast + push |
| Job Completion | Summary display; payment + tip info; rate the customer; navigate to earnings or home                                                                                                                                                                                                                                                                                                                                                                                                                                             |

#### Earnings, Messages & Profile

| Feature | Details |
|---------|---------|
|---------|---------|

|                              |                                                                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Earnings Dashboard</b>    | Total/weekly/monthly earnings; bar chart by day (Recharts); job breakdown with fee details; payout method display; payout history |
| <b>Messages</b>              | Conversation list with unread indicators; search by customer; encrypted chat modal; privacy name formatting ("FirstName L.")      |
| <b>Personal Info</b>         | Name, phone, full address editing                                                                                                 |
| <b>Vehicles</b>              | Add/edit/delete with pre-filled car database; make, model, year, color, plate                                                     |
| <b>Ratings &amp; Reviews</b> | Historical customer ratings; review text; pagination (10/page); average calculation                                               |
| <b>Tax Documents</b>         | Annual earnings summaries; gross, fees, tips, net breakdown; CSV export; 1099 income warning                                      |
| <b>Notifications</b>         | Push/system notification history; mark read; mark all read                                                                        |
| <b>Help &amp; Support</b>    | 5 help categories with articles; support ticket creation; reply threading                                                         |
| <b>Reporting</b>             | Job report, payout report, earnings report — all CSV exportable with date range filters                                           |
| <b>Account Security</b>      | Email update; password change; account deletion                                                                                   |
| <b>Explore</b>               | Nearby auto shops via Google Places; map + list view; distance filter                                                             |



## 4. Admin Dashboard Features

24 pages — Full platform operations, finance, support, analytics, and team management

### Dashboard & Operations

| Page          | Key Capabilities                                                                                                                                                                                        |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dashboard     | Real-time stats (active jobs, online providers, today's revenue, total users); 9-card platform management grid; recent alerts; pending provider queue; urgent tickets; real-time Supabase subscriptions |
| Jobs          | View all jobs (paginated); filter: All/Active/Completed/Cancelled; search; status badges with color coding; payment status; rating display                                                              |
| Live Dispatch | Real-time active job monitoring; auto-refresh 30s + Supabase subscriptions; stat cards (Active, Pending, En Route, In Progress); search by name/address                                                 |
| Services      | Create/edit services (name, description, base price, icon, status); toggle active/inactive; display order; job count per service                                                                        |

### User & Provider Management

| Page              | Key Capabilities                                                                                                                                                                                                                        |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Users             | All users (paginated, 15/page); filter: All/Customers/Providers/Admins/Suspended; expandable cards with job history; edit name/phone/role; suspend/unsuspend with audit log                                                             |
| Providers         | Detailed provider management; stats (earnings, jobs, rating, acceptance rate); vehicle/license info; services offered; document status; job + payout history; approve/reject/suspend/unsuspend/edit; email notifications on all actions |
| Provider Approval | Dedicated pending provider queue; document review with file previews; approve/reject individual documents with reasons; approve/reject entire application; email + in-app notification on all decisions                                 |
| Documents         | Document type configuration (create/edit required types); document review workflow: approve, reject, revoke; filter: All/Pending/Approved/Rejected; file download/preview                                                               |
| Team              | Admin team members; roles (Super Admin, Admin, Manager, Support); invite new members; suspend/unsuspend                                                                                                                                 |
| Directory         | Partner business directory; types: Repair Shop, Gas Station, Tow Yard, etc.; CRUD with geolocation; partner status toggle                                                                                                               |

### Finance & Payments

| Page     | Key Capabilities                                                                                     |
|----------|------------------------------------------------------------------------------------------------------|
| Payments | All job payments; filter by status: Paid/Unpaid/Failed/Refunded; search; CSV export; refund tracking |

|                |                                                                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Payouts        | Provider earnings + balances; jobs completed, earned, tips, fees, net owed, paid, balance; issue manual payouts; payout method display; CSV export               |
| Payout History | All processed payouts; filter: Paid/Processing/Pending/Failed; period dates, amounts; CSV export                                                                 |
| Finance Hub    | Revenue metrics; platform fee earnings; pending refund exposure; monthly revenue chart; job completion breakdown; payment status distribution; refund management |

Support, Analytics & Settings

| Page            | Key Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Support Tickets | All tickets; filter by status + priority; SLA tracking (urgent: 2h, standard: 24h); reply thread; assign to admin; change status/priority; admin notes                                                                                                                                                                                                                                                                                       |
| Notifications   | Compose broadcast notifications (title, message, type, audience); audience targeting: All/Customers/Providers; send history (last 50); recipient count                                                                                                                                                                                                                                                                                       |
| Analytics       | Real-time metrics; monthly revenue trend chart; job completion by status; service popularity; user growth; peak hours analysis                                                                                                                                                                                                                                                                                                               |
| Reporting Hub   | CSV exports for: Jobs, Refunds, Support Tickets, Audit Logs, Provider Profiles, Payout Records; date range filtering                                                                                                                                                                                                                                                                                                                         |
| Audit Trail     | All admin actions logged; filter by action/entity type; actor identification; details JSON; CSV export                                                                                                                                                                                                                                                                                                                                       |
| Settings        | <b>Fees:</b> cancellation %, commission %, service fee %, tax %, hazard fee \$, scheduling fee \$<br><b>Provider:</b> auto-approve toggle, document grace period, response timeout<br><b>Chat:</b> max message length, messages/page, retention days, image size/sharing<br><b>Platform:</b> maintenance mode, max job search radius, currency<br><b>Legal:</b> terms version/date, customer + provider terms templates, help center content |

## 5. Backend & Database

23 tables, 9 RPC functions, 15 triggers, 2 edge functions, 1 cron job

### Database Tables (23)

| Table                          | Purpose                                     | Key Columns                                                                        |
|--------------------------------|---------------------------------------------|------------------------------------------------------------------------------------|
| <b>jobs</b>                    | Service request records with full lifecycle | status, customer_id, provider_id, total_amount, payment_status, rating, timestamps |
| <b>profiles</b>                | User accounts (all roles)                   | email, name, phone, role, is_verified, status, rating                              |
| <b>provider_profiles</b>       | Extended provider data                      | is_online, is_verified, services[], rating, total_jobs                             |
| <b>documents</b>               | Provider document uploads                   | type, file_url, status, rejection_reason, expires_at                               |
| <b>document_types</b>          | Configurable document requirements          | name, is_required, is_active, display_order                                        |
| <b>services</b>                | Available service types                     | name, description, base_price, icon, is_active                                     |
| <b>vehicles</b>                | Customer vehicle information                | make, model, year, color, license_plate, is_primary                                |
| <b>chat_messages</b>           | In-job encrypted messaging                  | job_id, sender_id, sender_role, message, image_url                                 |
| <b>stripe_customers</b>        | Stripe customer ID mapping                  | user_id, stripe_customer_id                                                        |
| <b>payment_methods</b>         | Stored payment cards                        | stripe_payment_method_id, method_type, is_default                                  |
| <b>provider_payouts</b>        | Weekly payout records                       | total_earnings, tips, platform_fee, net_payout, status                             |
| <b>provider_payout_methods</b> | Provider bank/PayPal/Venmo                  | method_type, account_details, is_default                                           |
| <b>refunds</b>                 | Payment refund records                      | job_id, amount, reason, status                                                     |
| <b>support_tickets</b>         | Customer/provider support                   | subject, status, priority, assigned_to                                             |
| <b>ticket_replies</b>          | Support ticket threads                      | ticket_id, sender_role, message                                                    |
| <b>notifications</b>           | In-app notifications                        | user_id, type, title, message, action_url                                          |
| <b>device_tokens</b>           | Push notification tokens                    | platform, push_token, is_active                                                    |
| <b>push_notifications</b>      | Push delivery log                           | notification_type, status, sent_at                                                 |
| <b>broadcast_log</b>           | Admin broadcast history                     | title, audience, recipient_count                                                   |
| <b>platform_settings</b>       | Global config (key-value JSONB)             | key, value, updated_by                                                             |
| <b>provider_locations</b>      | Real-time GPS data                          | latitude, longitude, is_online                                                     |

|                    |                            |                                          |
|--------------------|----------------------------|------------------------------------------|
| job_events         | Immutable event audit log  | event_type, actor_id, metadata           |
| directory_listings | Partner business directory | business_name, type, lat/lng, is_partner |

## RPC Functions (9)

| Function                             | Purpose                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------|
| accept_job()                         | Race-safe job acceptance with SELECT...FOR UPDATE lock; emits pg_notify                  |
| cancel_job()                         | Authorized cancellation with actor verification; emits pg_notify                         |
| get_nearby_providers()               | Haversine geolocation dispatch; filters by distance, service, online status, active jobs |
| recalculate_provider_rating()        | Auto-recalculates average rating from completed jobs                                     |
| recalculate_customer_rating()        | Auto-recalculates customer rating from provider feedback                                 |
| send_broadcast_notification()        | Bulk notification send targeting audience segments                                       |
| upsert_device_token()                | Register or update device push notification token                                        |
| ensure_provider_setup()              | Guarantees profiles + provider_profiles rows exist                                       |
| suspend_expired_document_providers() | Daily auto-suspension of providers with expired required documents                       |

## Edge Functions (2)

### send-email

Sends templated transactional emails via Resend API. 7 templates: welcome, documents\_pending, document\_request, provider\_approved, provider\_suspended, customer\_invoice, provider\_completion.

Resend API

7 Templates

### create-payment-intent

Creates Stripe PaymentIntents for checkout. Gets/creates Stripe customer, attaches payment method, returns client secret for frontend confirmation.

Stripe

PCI Compliant

## Security: Row-Level Security (RLS)

All 23 tables have RLS enabled. Key policies:

- Jobs:** Users see own jobs; customers create; providers see pending; admins full access
- Profiles:** Anyone can view; self-update only; admins can update any
- Documents:** Providers manage own; admins manage all; public denied
- Payments:** Users manage own; service role processes
- Support:** Users see own tickets; admins manage all
- Settings:** Admins read/write; authenticated read public keys; anonymous read legal only

## 6. Email System

7 templates sent via Resend API through Supabase Edge Function

### Email Templates & Triggers

| Template            | Subject                                      | Triggered When                            | Sent To  |
|---------------------|----------------------------------------------|-------------------------------------------|----------|
| welcome             | Welcome to TORC!                             | Customer or provider signs up             | New user |
| documents_pending   | Documents Under Review                       | Provider submits documents                | Provider |
| document_request    | Action Required: Additional Documents Needed | Admin rejects document or requests update | Provider |
| provider_approved   | Your TORC Account is Approved!               | Admin approves provider application       | Provider |
| provider_suspended  | Account Suspended                            | Admin suspends provider account           | Provider |
| customer_invoice    | Service Complete — Invoice #XXXXX            | Job marked as completed                   | Customer |
| provider_completion | Job Complete — Earned \$XX.XX                | Job marked as completed                   | Provider |

### Email Architecture

**Provider:** Resend (<https://resend.com/>) • **From:** TORC <noreply@torcapp.com>  
**Method:** Supabase Edge Function receives { to, template, data } → renders HTML template → calls Resend API  
**Error handling:** All emails are fire-and-forget (non-blocking); errors logged but never block UI operations  
**Design:** Consistent branded HTML layout with teal gradient header, card-based content, responsive design

## 7. Real-Time Features

Supabase Broadcast, Postgres Changes, Presence, and Push Notifications

### Real-Time Capabilities

| Feature              | Technology              | Details                                                                                                                                                   |
|----------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPS Location Sharing | Supabase Broadcast      | Customer and provider exchange live location via <code>job-tracking-{jobId}</code> channel; includes heading + speed; auto-reconnect on visibility change |
| Encrypted Chat       | Supabase Broadcast + DB | AES-256-GCM encryption; PBKDF2-derived key from jobId; messages stored in <code>chat_messages</code> ; real-time delivery; image upload support           |

|                      |                             |                                                                                                                                           |
|----------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Audio Calling        | WebRTC + Supabase Signaling | Peer-to-peer audio via WebRTC; Supabase channel for ICE/SDP signaling; mute/speaker controls; call duration timer                         |
| Job Status Updates   | Postgres Changes            | Real-time subscription to jobs table changes; auto-updates UI on status transitions                                                       |
| Job Request Dispatch | Supabase Broadcast          | Customer broadcasts to provider-job-{providerId}; 30s timeout with audio/haptic; tiered radius expansion                                  |
| Chat Notifications   | Supabase Broadcast          | Cross-page notifications via chat-notify-{role}-{jobId}; unread badge on Message button; toast with decrypted preview                     |
| Push Notifications   | APNS / FCM                  | Native push via Capacitor; device tokens registered on login; delivered for: job accepted, provider arrived, job completed, job cancelled |
| Presence Tracking    | Supabase Presence           | Track online/offline status of peers during active jobs; connection state indicator                                                       |
| Admin Dashboard      | Postgres Changes            | Real-time updates to dashboard metrics: jobs, profiles, provider_profiles, support_tickets, refunds                                       |

## Key User Flows Summary

### Customer Service Request

Home → Who Needs Help → Confirm Location → Select Service → Schedule → Pay → Match → Track → Rate

8-step flow

Stripe payment

### Provider Job Execution

Go Online → Receive Request (30s) → Accept → Navigate → Arrive → Start Service → Take Photos → Complete

30s timeout

GPS + Chat

### Provider Onboarding

Signup → Verify Email → Select Services → Upload Documents → Setup Payout → Await Admin Approval

Admin review

Email notifications

### Admin Provider Approval

Review documents → Approve/Reject individual docs → Approve/Reject application → Email + notification sent

Role-restricted

Audit logged